

ALESSANDRO MARTINOLLI

BACKEND SOFTWARE ENGINEER

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PROFILE

Backend software engineer with a double Master's in Computer Science and a year building event-driven, Kafka-based systems in travel tech at Amadeus. Comfortable owning services in production, from development through deployment and incident response. Strong collaborator across cultures and teams, with a growing focus on AI, system reliability, and technical leadership.

WORK EXPERIENCE

Software Engineer Consultant — Amadeus (via Audensiel), France Jul 2025 – Present

- Backend developer on large-scale, event-driven systems within the OFMS platform, building and maintaining Java microservices in production.
- Design and operate Kafka-based event-driven services, handling message processing, reconciliation flows, and inter-service communication at scale.
- Collaborated with production maintenance: monitoring, incident response, and deployment / pod handling across Kubernetes environments.
- Diagnosed and resolved a production event-loop thread-blocking incident in a core connector service, and documented the root cause for the wider team.
- Built reusable internal tooling and strengthened test coverage with parameterised JUnit 5 suites and JMH benchmarking.
- Mentored new joiners, led cross-team collaboration and task prioritisation, and contributed to the team's technical practices under a SAFe framework.

Java · Kafka · Kubernetes · JUnit 5 · JMH · SAFe · Jira

Implementation Engineer — OpenWay (Remote) Apr 2025 – Jul 2025

- Configured and customised the Way4 platform for card-issuing projects: card management, transaction processing, and customer onboarding.
- Supported integrations with external systems and ensured alignment with Visa/Mastercard standards.

Java · SQL · Bash

Software Developer (Internship) — ESTECO, Trieste, Italy May 2024 – Apr 2025

- Built a library for energy and runtime profiling in Julia, applied to large sparse linear systems.
- Optimised the company's code for speedups of up to 3× on CPU and 10× on GPU.

Julia · C++ · CUDA · NVIDIA Nsight · Python · Linux

Research Assistant — University of Illinois Chicago, USA Nov 2023 – May 2024

- Studied NVIDIA GPU architecture and energy-optimisation techniques; implemented predictive models to reduce energy consumption on heterogeneous systems.
- Used profilers and OpenMPI to analyse and tune performance across CPU/GPU workloads.

Python · C++ · CUDA · NVIDIA Nsight · OpenMPI · Linux

Web Developer — PoliMi Sailing Team, Milan 2022

- Led the IT section of the Logistics Department.

EDUCATION

Politecnico di Milano — M.Sc. Computer Science Engineering (HPC profile) 2022 – 2025

University of Illinois Chicago — M.Sc. Computer Science 2023 – 2025

Politecnico di Milano — B.Sc. Computer Science Engineering 2019 – 2022

SELECTED PROJECTS

AI Code Assistant (concept) — VS Code extension that detects data-structure / algorithm patterns and suggests annotated alternatives (time & space complexity).

Distributed Computing for Cloud — graph analysis with Apache Spark & MapReduce on Scala.

HPC / GPU Project — CUDA with shared memory over the Tartan benchmark and grCUDA.

BeCrew — sailing community platform connecting boat owners with crew (side project).

Software Engineering Project — client-server Java application.

SKILLS & LANGUAGES

Technical: Java · Kafka · Kubernetes · Event-Driven Architecture · Microservices · SQL · Python · C++ · CUDA · Bash · Linux · Git · SAFe · Jira

Languages: Italian (native) · English (fluent) · French (professional, developing)

Interests: Sailing · building side projects (web, AI & developer tooling) · entrepreneurship · open to business travel.